AWS is built for failure, even in the rare case that an entire data center fails. The AWS regions consist of multiple data centers grouped into availability zones. By distributing your workload among multiple availability zones, you are able to recover from an data center outage. Two pitfalls when building a highly available setup over multiple availability zones follow:

- Data stored on network-attached storage (EBS) won't be available after failing over to another availability zone by default. In this case, you can end up having no access to your data (stored on EBS volumes) until the availability zone is back online (but you won't lose your data in this case).
- You can't start a new virtual machine in another availability zone with the same private IP address. That's because subnets are bound to availability zones, and each subnet has a unique IP address range. By default, you can't keep the same public IP address automatically after a recovery, as was the case in the previous section with a CloudWatch alarm triggering a recovery.

We will deal with those pitfalls at the end of this section. In this section, you'll improve the Jenkins setup from the previous section, add the ability to recover from an outage of an entire availability zone, and work around the pitfalls afterward.

13.2.1 Availability zones: Groups of isolated data centers

As you've learned, AWS operates multiple locations worldwide, called *regions*. You've used the region US East (N. Virginia), also called us-east-1, if you've followed the examples so far. In total, there are 23 publicly available regions throughout North America, South America, Europe, Africa, and Asia Pacific.

Each region consists of multiple availability zones (AZs). You can think of an AZ as an isolated group of data centers and a region as an area where multiple availability zones are located at a sufficient distance. The region us-east-1 consists of six availability zones (us-east-1a to us-east-1f), for example. The availability zone us-east-1a could be one data center or many. We don't know because AWS doesn't make information about their data centers publicly